CaseLink

Project Team

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Chapter 1

Introduction

This mid evaluation document lists the requirements, design artifacts and high level system diagrams for CaseLink. CaseLink is an AI-based software solution for law firms. It provides a smart solution that integrates state-of-the-art features, including large language models (LLMs) and generative AI, into a modular platform to streamline a law firm's inner processes. 21% of law firms that utilised legal software reported a larger volume of case work than those that did not implement such solutions [1]. Utilising the power of the cloud and online databases, we hope to deliver a product that can boost law firm efficiency, help ease the entire process for both the client and the legal team, and assist lawyers in preparing to deal with cases.

1.1 Problem Statement

The legal professionals, particularly law firms and lawyers in Pakistan face various challenges e.g managing documentation for each case, scheduling case hearing dates, generating legal documents that follow certain formats and preparing strong arguments based on latest legal data. Law firms in Pakistan use social media groups like WhatsApp for document and template sharing. Firm's lawyers have to manually sift through a myriad of articles and judgements for preparing arguments. Firm's clerks need to update lawyers on next hearing dates assigned to ensure their availability in courts. All of the mentioned workflows can be automated to save time, effort and resources of a law firm.

CaseLink is an all in one software solution meant to utilize the power of automation and AI to support law firms. The software will be designed to address the challenges of law firms by automating routine tasks and incorporating AI to help their lawyers. With CaseLink, law firms can achieve a significant boost in performance and efficiency by saving lawyer's precious time through automation and AI-driven solutions.

1.2 Scope

CaseLink is a web based application for law firms to manage their cases and clients as well as derive useful insights from legal documents and judgements to build strong cases. There are 4 main users (stakeholders) that CaseLink supports i.e. firm managers, lawyers (associates), clerks and clients.

CaseLink allows the firm managers to register cases and clients. They can manage all aspects of client and case data, assign cases or hearing dates to their associates with help from CaseLink's recommendation system. Firm managers as well as lawyers can generate legal documents using CaseLink's document generation module. Lawyers can view their calendar and receive notifications on important case dates. They can use CaseLink's advisory AI to extract relevant judgments for their cases. Clerks have minimal authorization (e.g. updating hearing dates for any case or uploading case relevant information) that is required by the firm manager. Clients can interact with the lawyers using in-app chat facility to provide documents and receive case updates from lawyers.

1.3 Literature Review

While performing market analysis for sofwares and tools with similar fucntionalities that CaseLink offers, we found software solutions like clio, mycase and leap providing case management and scheduling services as well as AI tools like lawgro for advise on legal proceedings. With CaseLink, we aim to provide an integrated solution that combines management aspects with automation and AI.

CaseLink can fill the gap in the availability of an all-encompassing platform that not only addresses client inquiries but also supports the back-end needs of law firms. Specifically, law firms would benefit from a system that integrates AI-powered document generation, case management, billing, and client communication into one cohesive platform.

Following table provides a market analysis breakdown of CaseLink.

	Case Management	Task & Calender	In-App Chat	Document Generation	AI Assistant
CL	✓	✓	✓	✓	✓
Clio	✓	✓	✗	✓	✗
mycase	✓	✓	✓	✗	✗
LEAP	✓	✓	✗	✗	✓
LAWGRO	✓	✓	✗	✗	✗
	✓	✓	✓	✗	✗

Figure 1.1: CaseLink's Market Analysis

1.4 Modules

1.4.1 Case Management

It involves the storage of client and case data for ease of access by the relevant lawyers.

1. Case tab: Displays all the registered cases in a list.
2. Case card: Contains a specific case's description, relevant documents and other details.
3. Client details: Displays all the clients registered with the firm alongside their cases, status and other details.

1.4.2 Lawyer Hub

Manages Lawyer Persona and provide smart case assignment suggestions.

1. Lawyer info tracker: Keeps track of lawyer's case history and skills.
2. Smart suggestion: Recommend lawyers to firm managers which are better suited to tackling certain cases.

3. Case assignment: Allows firm managers to assign cases or hearing dates to lawyers.

1.4.3 Scheduling and Notification

Update lawyer schedules for assigned cases and send notifications to the relevant users.

1. Calendar: Displays cases and dates assigned on a calendar view for each lawyer.

2. Notification: System generates notification to inform relevant users before hearing dates.

1.4.4 Legal Convo

In-app chat facility for communication between lawyers and clients alongside functionality for sharing media like case documents.

1. Chat app: Allows for interaction between lawyers and clients.

2. Document support: Allows users to share documents over chat.

1.4.5 DocuCraft

Generates properly formatted legal documents as per the necessary requirements and specifications.

1. Template support: Allows user to choose type of document for generation.

2. Document generation: Generates document based on details provided by the user.

1.4.6 LLM Based Legal Assistant

AI chatbot that helps lawyers build arguments by providing relevant judgments for criminal cases and help analyze legal documents.

1. Document insights: Analyzes documents provided by user for specific details.

2. Advisory AI: Helps user build arguments for criminal cases based on judgments of previous similar cases.

1.5 User Classes and Characteristics

Table 1.1: CaseLink Users and Characteristics

User Class	Description
Firm Manager	• Lead the entire law firm. • Responsible for filing and assigning cases. • Build arguments for cases. • Create, approve and upload legal documents. • Appear in court for legal proceedings.
Lawyer (Associate)	• Work on cases and arguments assigned to them. • Create and upload legal documents and build arguments for cases. • Appear in court for legal proceedings.
Clerk	• Keep track of case dates for cases. • Update certain case details, e.g., next date and order.
Client	• Contact law firms for filing or defending cases. • Need to be updated regarding their case status. • Provide relevant case documents required by the law firm.

Chapter 2

Project Requirements

This section presents the use-case diagram for CaseLink along with a formal description of its functional and non-functional requirements, derived from the requirement-gathering phase. These requirements provide a clear understanding of the system's expected behavior and performance standards.

Additionally, this section includes a comprehensive domain model for CaseLink, which outlines the system's key entities and relationships. Detailed descriptions of the targeted use cases for the first iteration of development are also provided in this section to guide the implementation process. All the diagrams in this section are built using "draw.io" tool [2].

2.1 Use-cases

2.1.1 Use-case Diagram

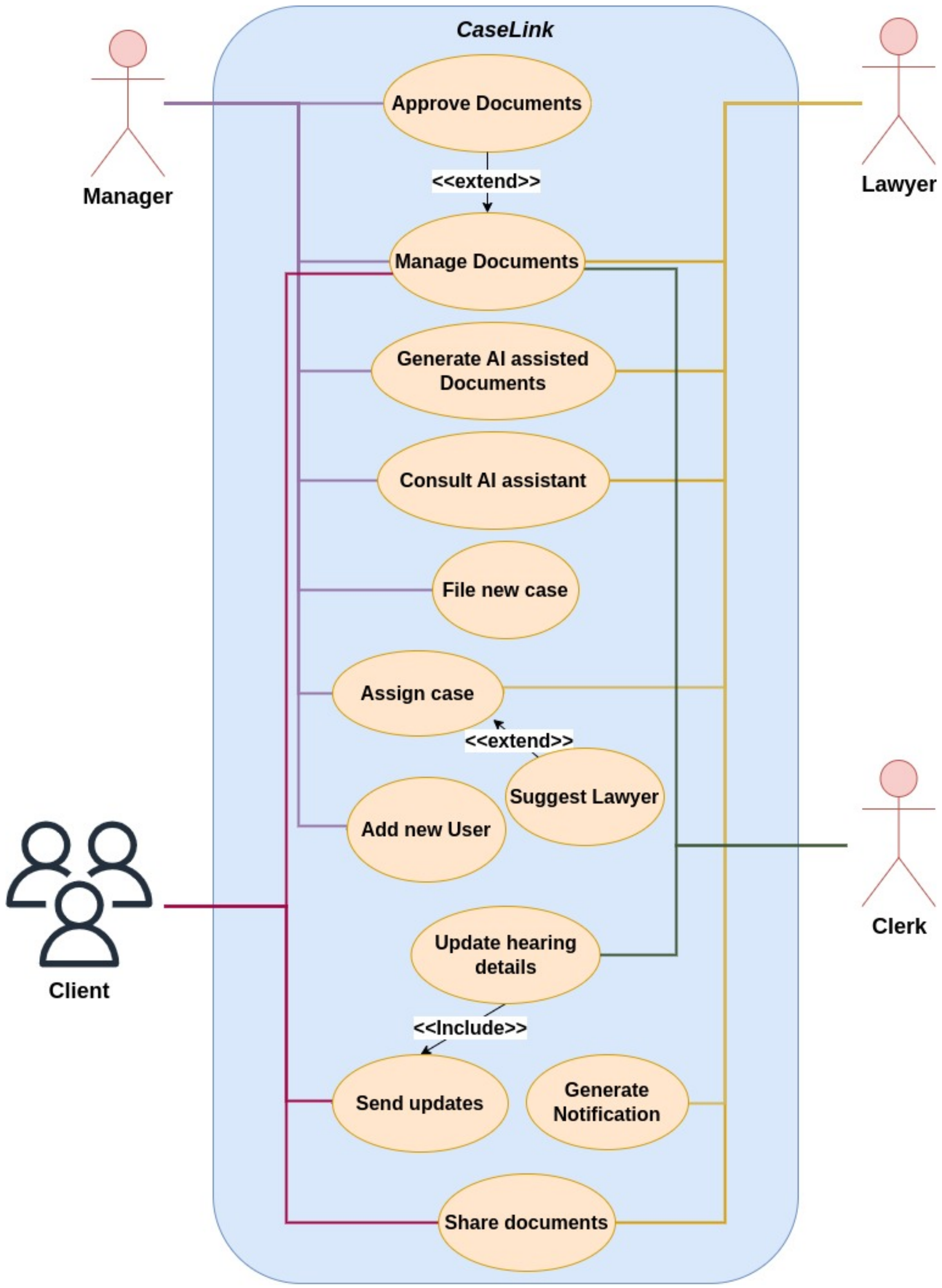

Figure 2.1: CaseLink Use-case Diagram

2.1.2 Iteration 1 Use-cases

Following are the extended use-cases for the module that is included in the first iteration i.e. Case Management.

ID	1.1
Assignee	Abtaal Aatif
Module	Case Management
Name	Manage Documents
Description	Users can upload and remove documents.
Primary Actor(s)	Manager, Lawyer, Clerk
Preconditions	1. The user has successfully logged in and is fully authenticated.
Basic Flow	Step 1: The user navigates to the cases page and opens a particular case. Step 2: All case-relevant documents are displayed. Step 3: Based on role: **Clerk**: Can upload case-relevant documents and request approval. **Lawyer**: Can view, upload, and remove documents. **Manager**: Can upload, remove, and approve documents. Step 4: Click the save changes button. Step 5: The system tracks changes like document uploading/removal.
Alternate Flow	3.1: Invalid Document Format: Solution: System prompts user to upload valid document format. 4.1: No changes made: Solution: The system takes no action.
Post Condition	1. In case of upload, the document is stored. 2. In case of removal, the document is deleted. 3. Approved documents stay in the system; disapproved ones are removed, and the uploader is notified.

Table 2.1: Use-case 1.1: Manage Documents

ID	1.2
Assignee	Abtaal Aatif
Module	Case Management
Name	Approve Documents
Description	Managers can approve or disapprove of documents uploaded by clerks and lawyers.
Primary Actor(s)	Manager
Preconditions	1. The manager is authenticated. 2. There is at least one document for approval.
Basic Flow	Step 1: The manager navigates to the documents tab. Step 2: Documents requiring approval are displayed with approve/disapprove options. In case of approval: The uploader is notified, and the document remains in the system. In case of disapproval: The uploader is notified with comments, and the document is removed.
Alternate Flow	2.1: Document status not updated: Solution: The system prompts manager to try again.
Post Condition	1. The sender is notified with comments (if any) about the approval/disapproval of document.

Table 2.2: Use-case 1.2: Approve Documents

ID	1.3
Assignee	Kalsoom Tariq
Module	Case Management
Name	File new case
Description	The system shall allow the manager to file a new case on behalf of their client.
Primary Actor(s)	Manager
Preconditions	1. The manager is authenticated. 2. Basic client and case information required for filing a case is available.
Basic Flow	Step 1: They navigate to the File Case Page. Step 2: Enter the client and case details. Step 3: Upload the case filing document. Step 4: Click on “File Case” Button.
Alternate Flow	2.1: Details required for filing are missing: Solution: The manager acquires the details and continues. 3.1: Wrong document format: Solution: The system notifies the manager to upload the correct format document. 4.1: The API call returns failure: Solution: The manager tries again from Step 1 with an active internet connection.
Post Condition	1. The filed case appears on the case list. 2. The manager and lawyer can see the filed case.

Table 2.3: Use-case 1.3: File New Case

ID	1.4
Assignee	Kalsoom Tariq
Module	Case Management
Name	Add new user
Description	The system shall allow the manager to add new lawyers and clerks to the firm.
Primary Actor(s)	Manager
Preconditions	1. The manager has successfully registered a firm on CaseLink. 2. The manager is authenticated.
Basic Flow	Step 1: The manager navigates to Add User page. Step 2: Select the type of user to add i.e., clerk or lawyer. Step 3: Provide the user's name and enter their valid email in the field. Step 4: Click on Add User button.
Alternate Flow	3.1: The email already exists in database: Solution: The system notifies the manager that a user already exists with this email and prompts them to try again. 4.1: The API call returns failure: Solution: The manager tries again from Step 1 with an active internet connection.
Post Condition	1. The user receives an email containing login credentials. 2. The user successfully logs in to CaseLink. 3. The manager can see the new user in the Registered Users list.

Table 2.4: Use-case 1.4: Add New User

2.2 Functional Requirements

2.2.1 Case Management

1. The system shall provide a daily case list view, including case title, FIR number, and client name.
2. Users shall be able to check the status of a specific case, including:
    - Judge information
    - Court information
    - Advocates involved

- Case description
- Next date

3. The system shall support storage and retrieval of:
    - Evidence copies (typed or handwritten)
    - Order sheets (records of each case proceeding)
    - Plaints and Written statements (or FIR in criminal cases)
4. The system shall categorize cases into Criminal, Civil, and Family types.
5. The system shall store client information, including:
    - Name
    - Address
    - Contact number / Email
6. The system shall allow managers to decide whether to profile each client.

2.2.2 Lawyer Hub

1. The system shall support case assignment to Lawyers (Associates).
2. The system shall support the following user roles with specified access levels:
    - Firm Manager: Full case read and write access
    - Lawyer (Associates): Full case read and write access
    - Clerks: Access granted by the firm manager
    - Clients: Access granted by the firm manager
3. The system shall generate reports on:
    - Number of new cases added
    - Number of cases decided
    - Outcome of decided cases
    - Lawyer-wise case analytics

2.2.3 Scheduling and Notification

1. The system shall send email notifications to clients and assigned lawyers for case appearances, including date, case number, title, and court name.

2.2.4 Legal Convo

1. The system shall facilitate communication with clients, including sending legal updates and requests for information.

2.2.5 DocuCraft

1. The system shall provide document generation capabilities from templates (to be provided by the firm manager).
2. The system shall include an approval workflow for document filing, requiring validation from the firm manager.

2.2.6 LLM Based Legal Assistant

1. The system shall provide document summarization capabilities.
2. The system shall assist in argument preparation by referencing previous relevant case judgements.

2.3 Non-Functional Requirements

2.3.1 Reliability and Availability

- The system shall maintain 99.9% uptime.
- The system shall implement automatic backups and disaster recovery plans.
- The system shall support offline functionality in future iterations.

2.3.2 Accessibility

- The system shall be primarily accessible via desktop and mobile devices.
- The system shall be implemented as a responsive web application, optimized for both desktop and mobile use.
- The system shall be fully functional and optimized for the following web browsers:
  - Google Chrome
  - Mozilla Firefox
  - Microsoft Edge

2.3.3 Scalability

- The system shall support a minimum of 20 concurrent users without degradation in performance.

2.3.4 Performance

- The system shall respond to most actions (e.g., loading a case, retrieving documents) within 4-5 seconds over a 20 Mbps or faster Internet connection.

2.3.5 Security

- The system shall implement Single Sign-On (SSO) for user authentication.
- The system shall lock a user account after 5 failed login attempts within a 15-minute period.
- The system shall treat all documents as publicly accessible, as per the firm's policy.

2.4 Domain Model

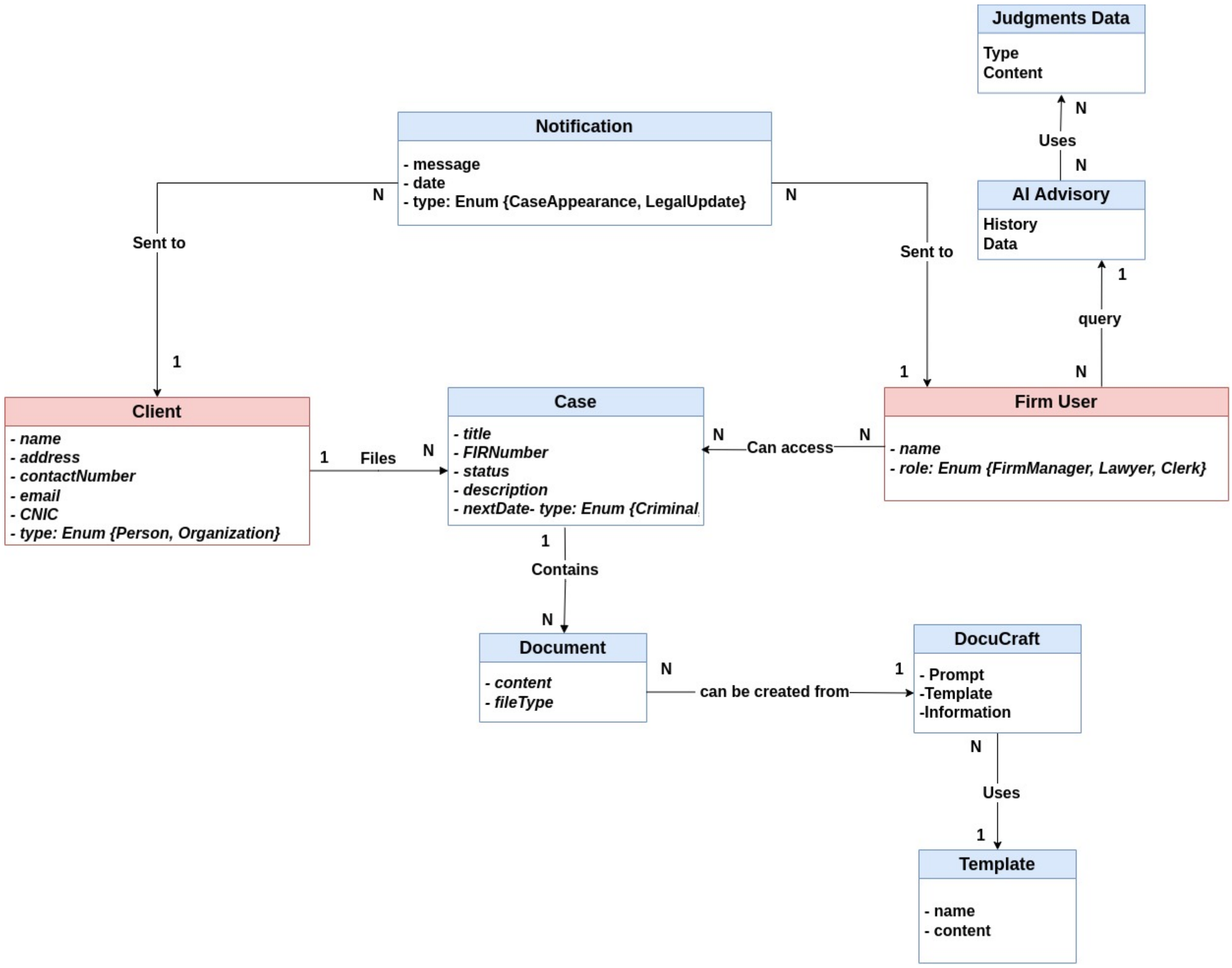

Figure 2.2: CaseLink's Domain Model

Chapter 3

System Overview

3.1 Architectural Design

This section provides an in-depth overview of the architectural design for CaseLink system. It includes system's high-level architecture diagram, class diagram that presents the object-oriented relationships. Furthermore, it also includes activity and sequence diagrams representing the workflows of key use-cases in case management module and state transition diagrams for critical states of the system. Lastly, CaseLink's data design model is built using a JSON schema to define the its data structures.

3.1.1 Architecture Diagram

The architecture follows a Microservices-based pattern. The frontend and backend are decoupled, with services like case management, document storage, and legal assistant functions offered through REST APIs. This modular structure allows each backend service (e.g., document generation, LLM insights) to scale independently, while APIs act as the central communication layer.

The architecture diagram given below is built using "figma" tool [3]. This architecture leverages a web-based frontend (built using NextJs) connected to a microservices-oriented backend. The backend services are responsible for managing case data, facilitating communication between lawyers and clients, generating legal documents, and providing AI-driven legal assistance. MongoDB stores structured case and client data, AWS S3 handles file storage, and Redis acts as a vector database for fast data retrieval. Machine learning models support smart recommendations for lawyer assignments and legal document analysis. The architecture is scalable, flexible, and optimized for responsiveness across devices [4].

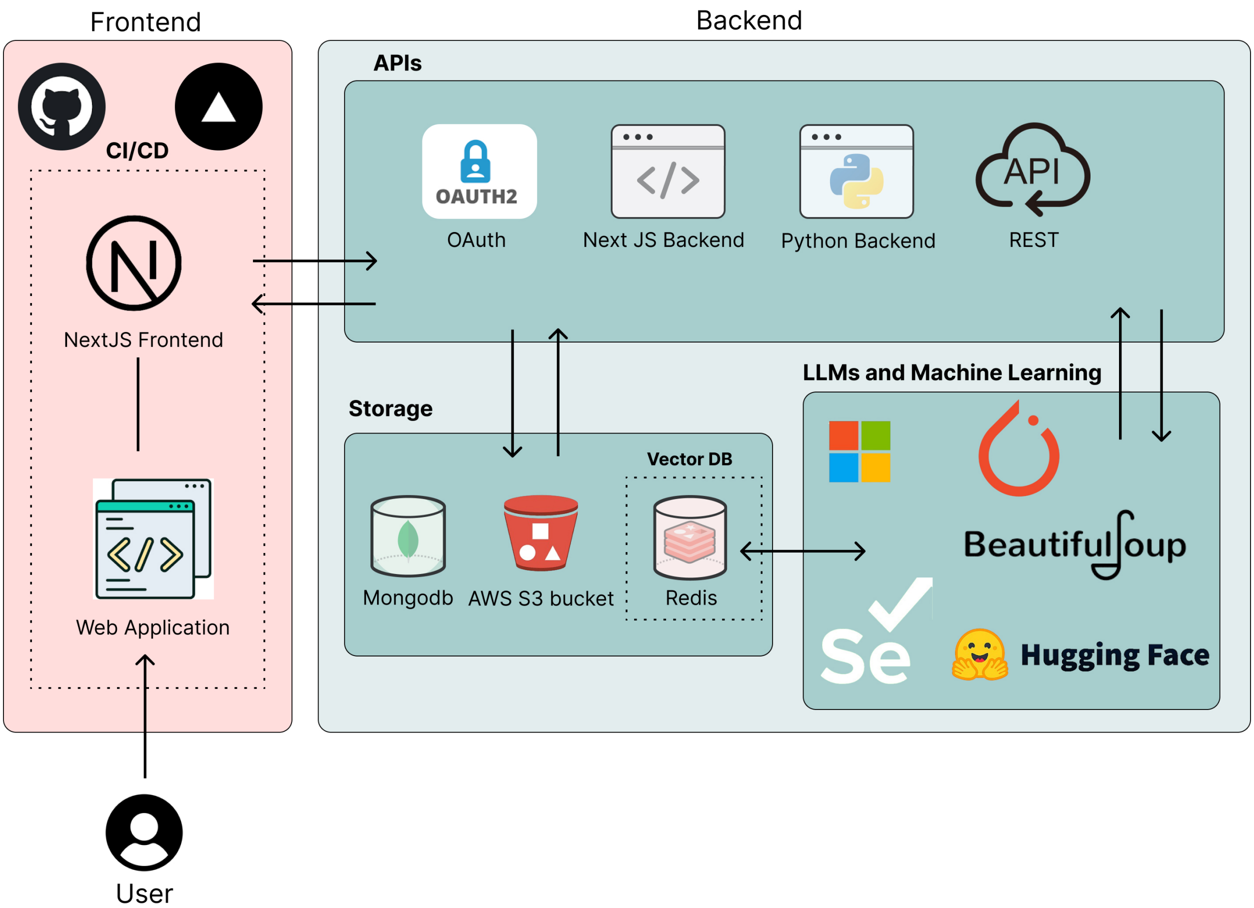

Figure 3.1: CaseLink's Architecture Diagram

3.1.2 Class Diagram

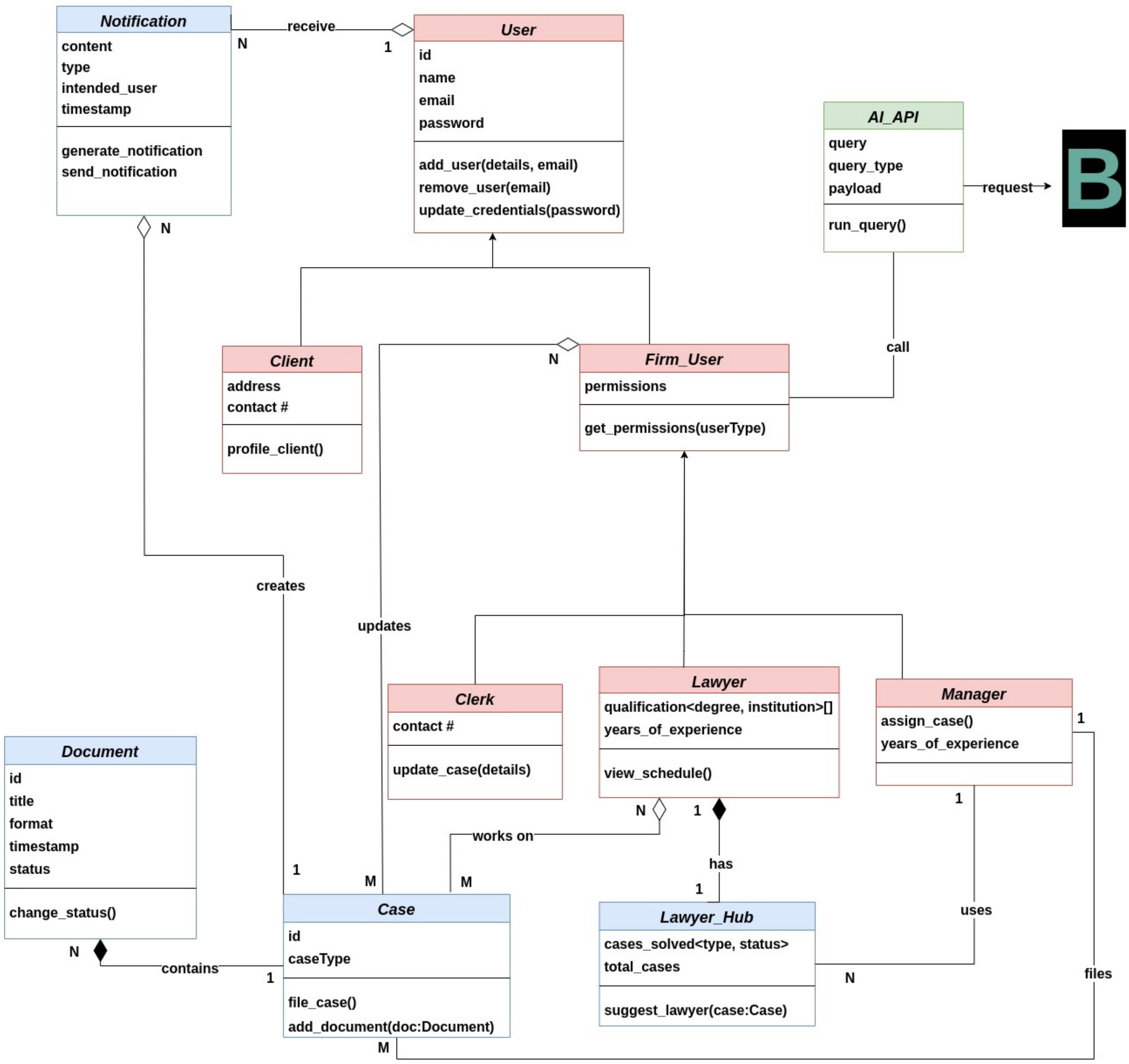

Figure 3.2: CaseLink's Class Diagram Block A

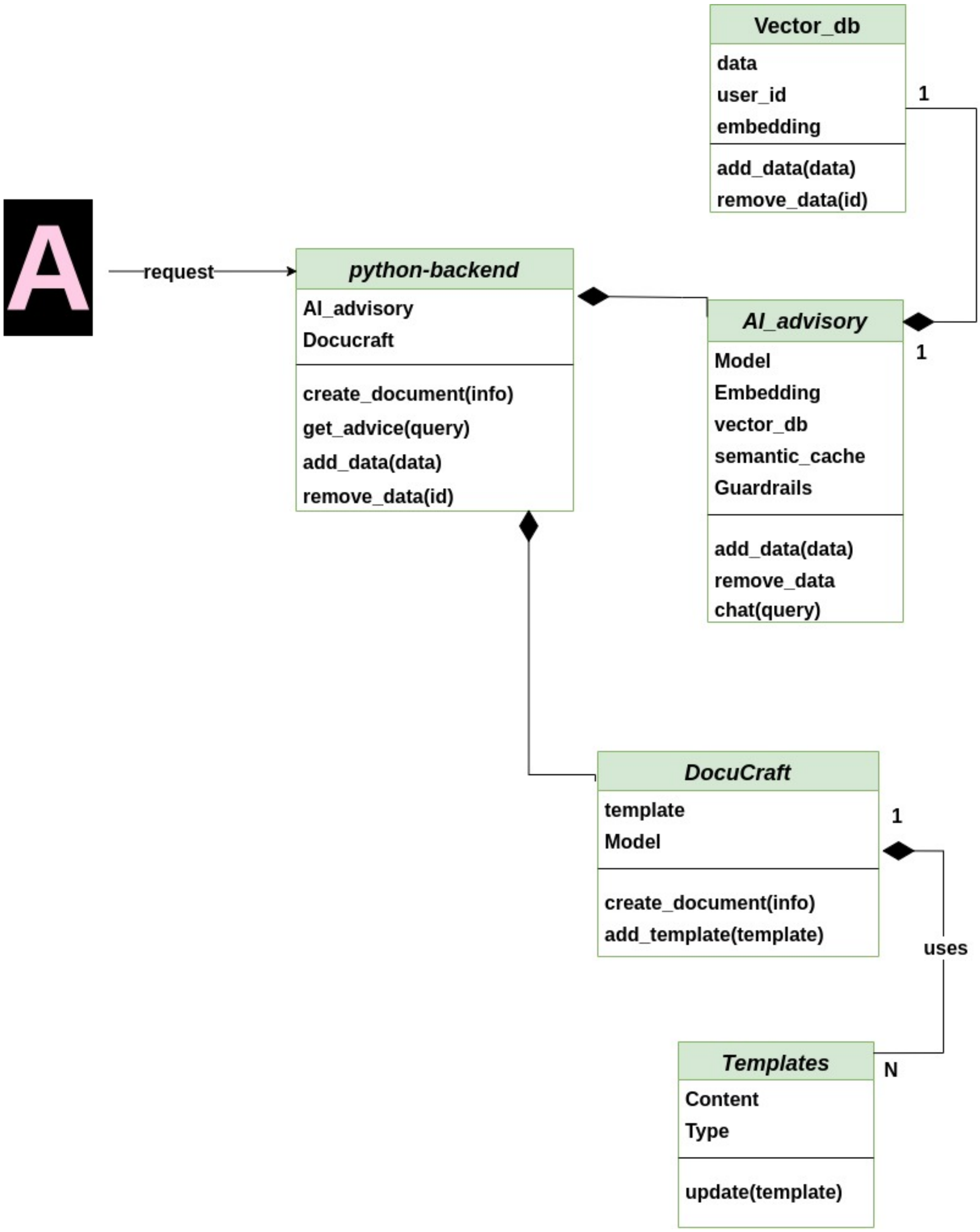

Figure 3.3: CaseLink's Class Diagram Block B

3.2 Design Models

3.2.1 Activity Diagrams

The following activity diagrams illustrate the step-by-step workflows of the key use-cases in case management module of CaseLink, focusing on the interaction between users and the system. These diagrams represent the logical sequence of actions performed by different actors, such as managers, lawyers, and clerks, to accomplish tasks like document management, case filing, and user addition. These diagrams provide a clear understanding of how the system handles these operations [6].

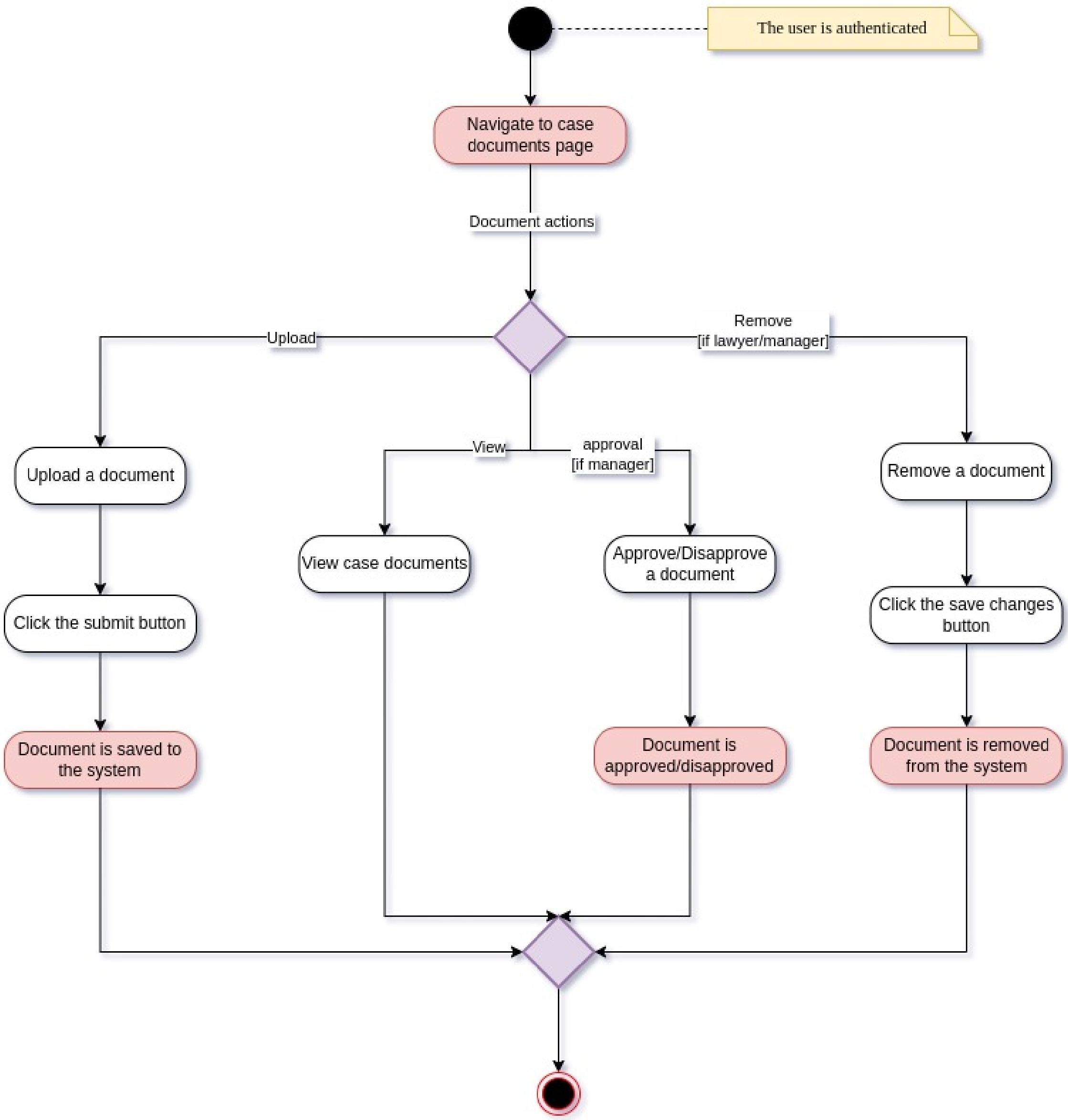

Figure 3.4: Activity Diagram - Use-case 1.1: Manage Documents

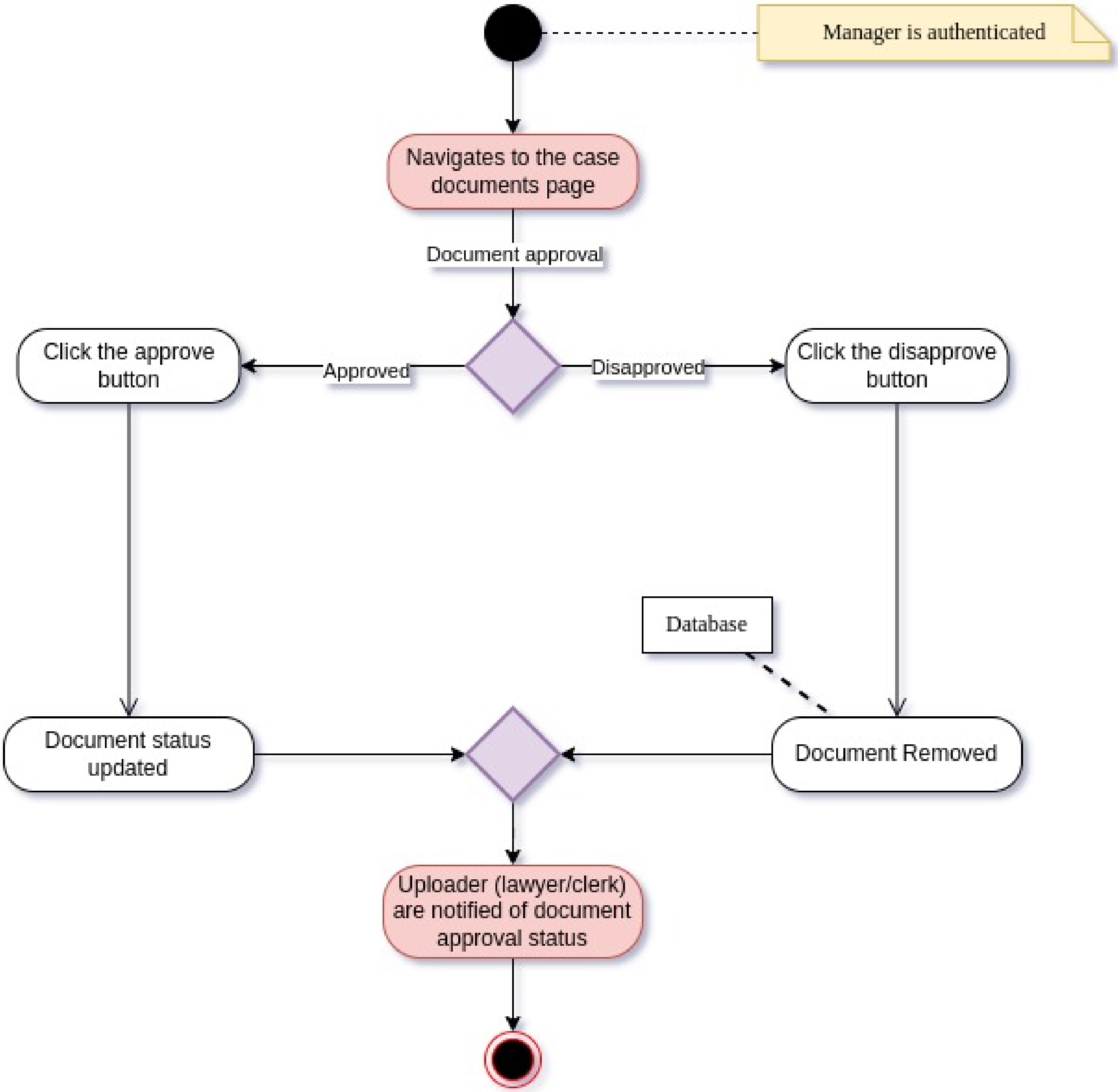

Figure 3.5: Activity Diagram - Use-case 1.2: Approve Documents

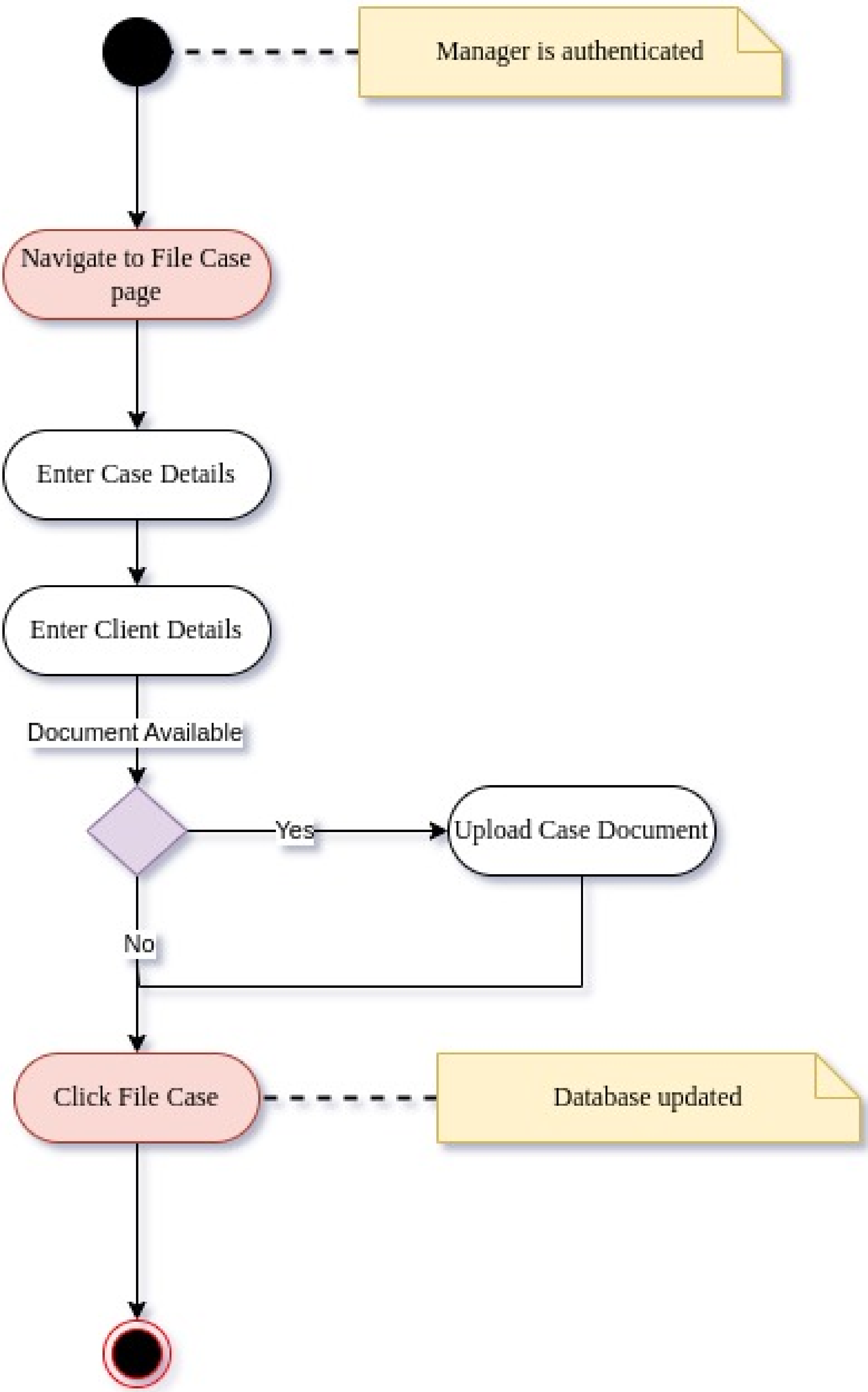

Figure 3.6: Activity Diagram - Use-case 1.3: File New Case

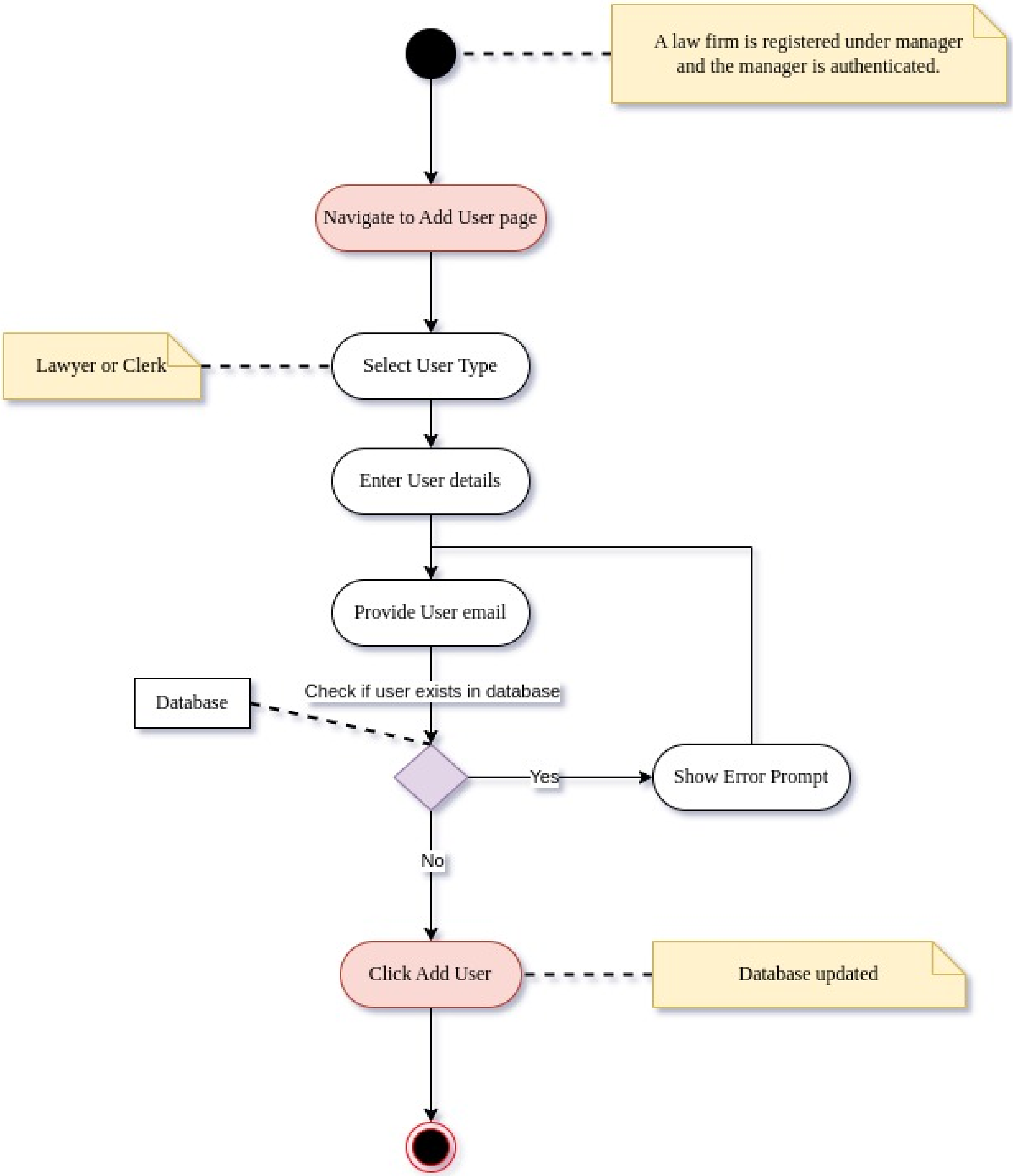

Figure 3.7: Activity Diagram - Use-case 1.4: Add New User

3.2.2 Sequence Diagrams

The sequence diagrams (SDs) presented below capture the dynamic interaction between users and the system during the execution of use-cases in case management module. Each diagram highlights the sequence of messages exchanged between actors and the system, detailing how input is processed and the corresponding system responses. These SDs provide an understanding of how specific user actions are translated into system operations [7].

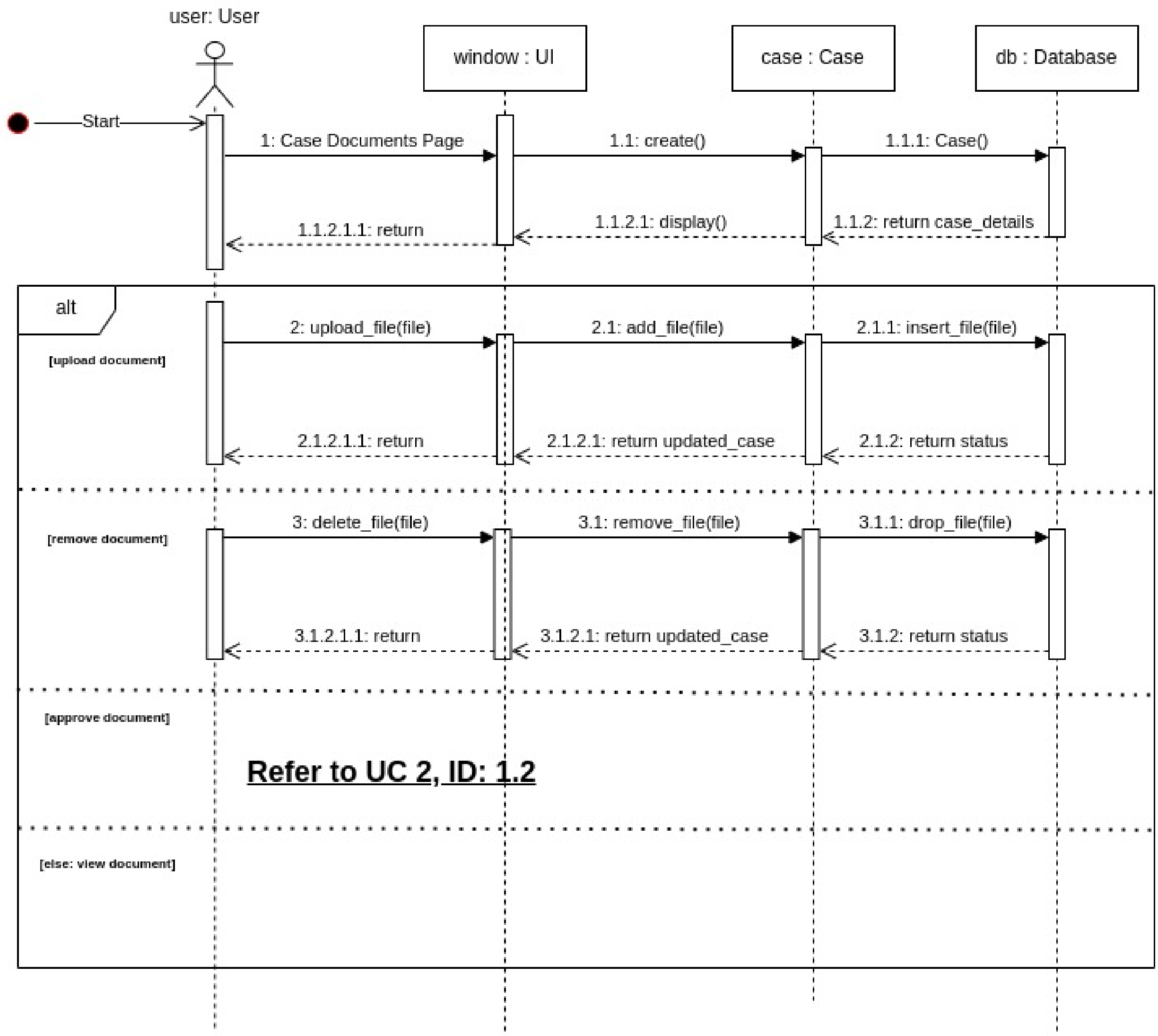

Figure 3.8: System Sequence Diagram - Use-case 1.1: Manage Documents

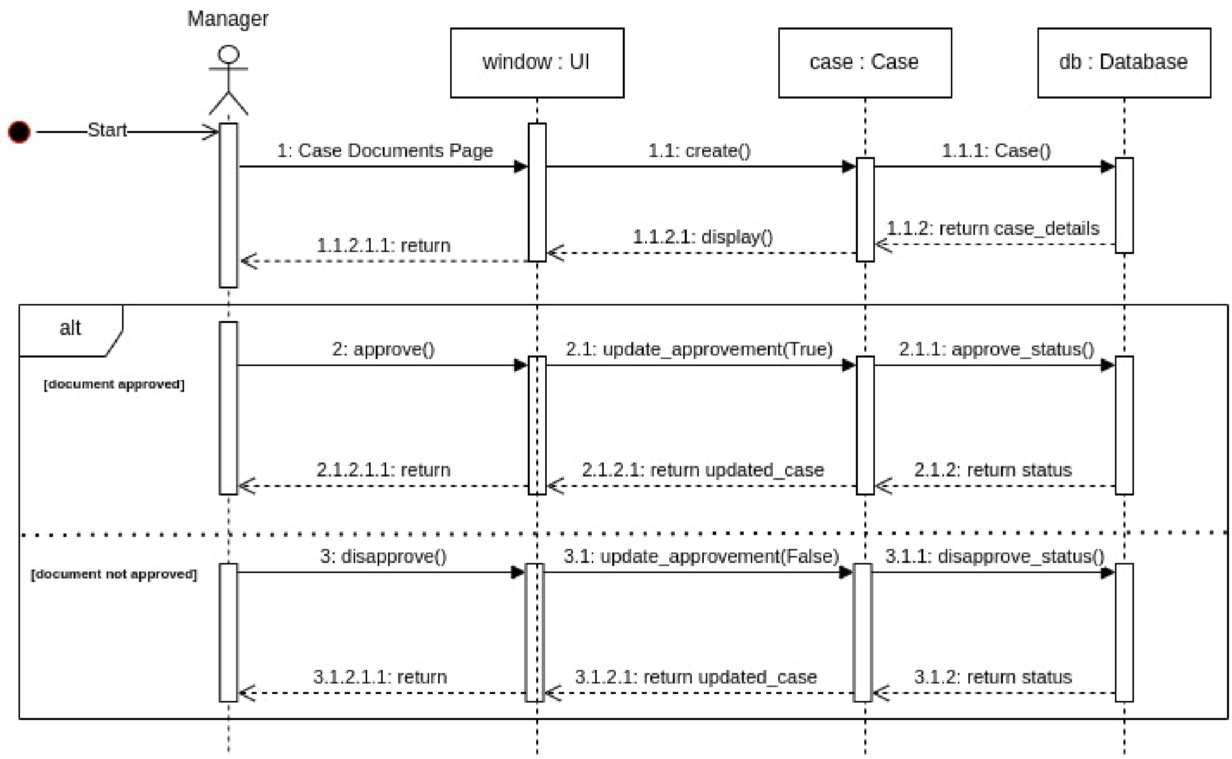

Figure 3.9: System Sequence Diagram - Use-case 1.2: Approve Documents

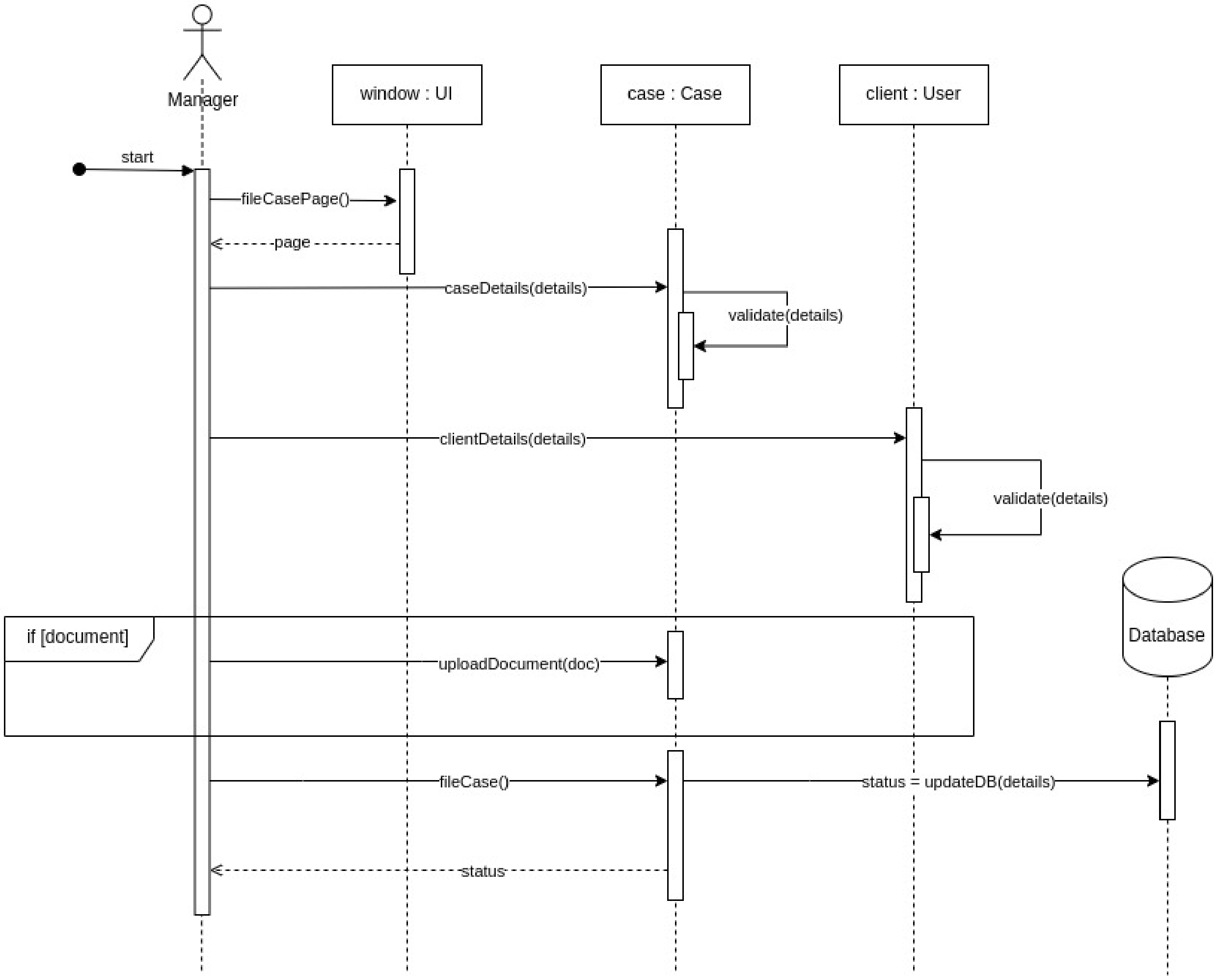

Figure 3.10: System Sequence Diagram - Use-case 1.3: File New Case

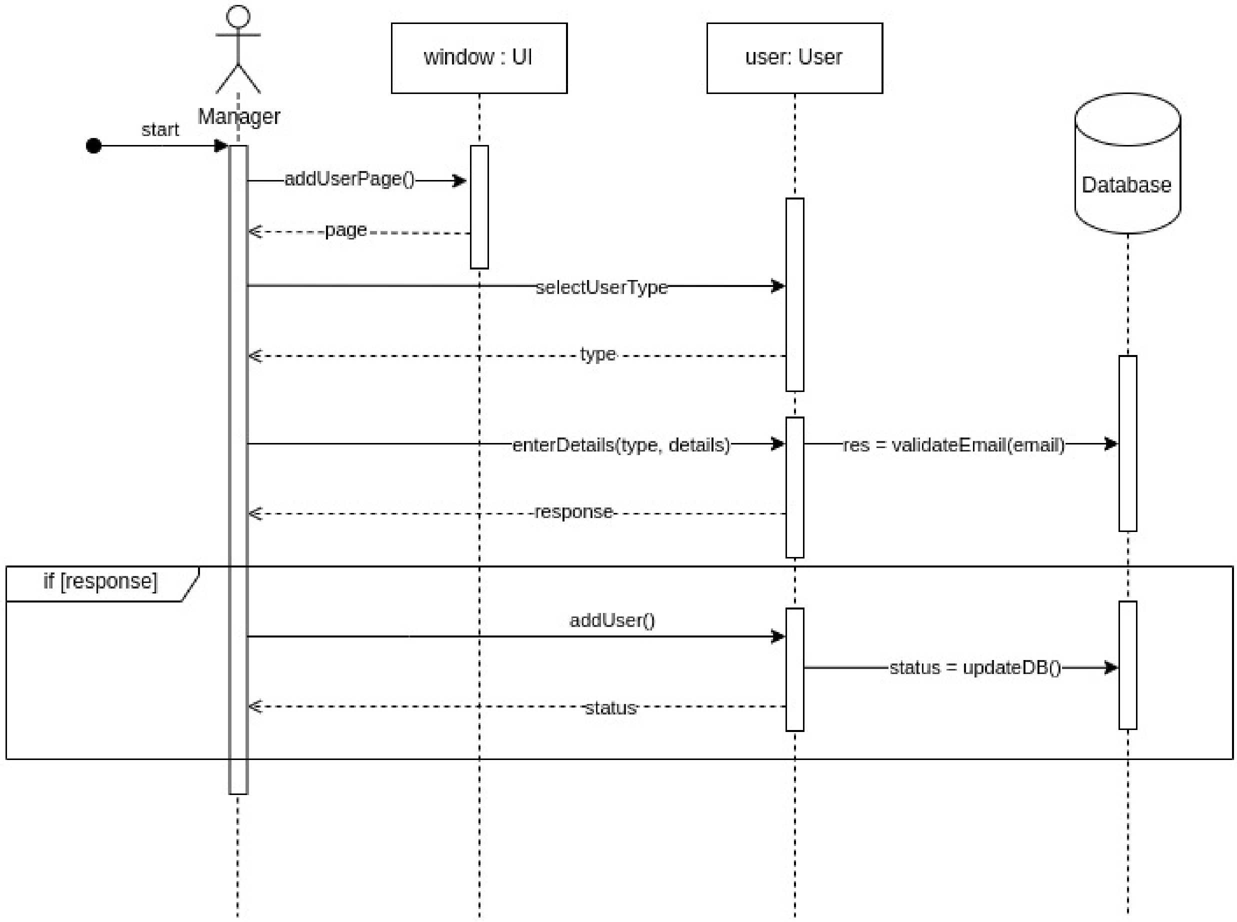

Figure 3.11: System Sequence Diagram - Use-case 1.4: Add New User

3.2.3 State Transition Diagram

Following state transition diagrams are made only for use-case 1.1 and 1.2 as these use-cases of case management module provide a clear overview of the lifecycle of documents and a better understanding of the processes involved in managing and approving legal documentation [5].

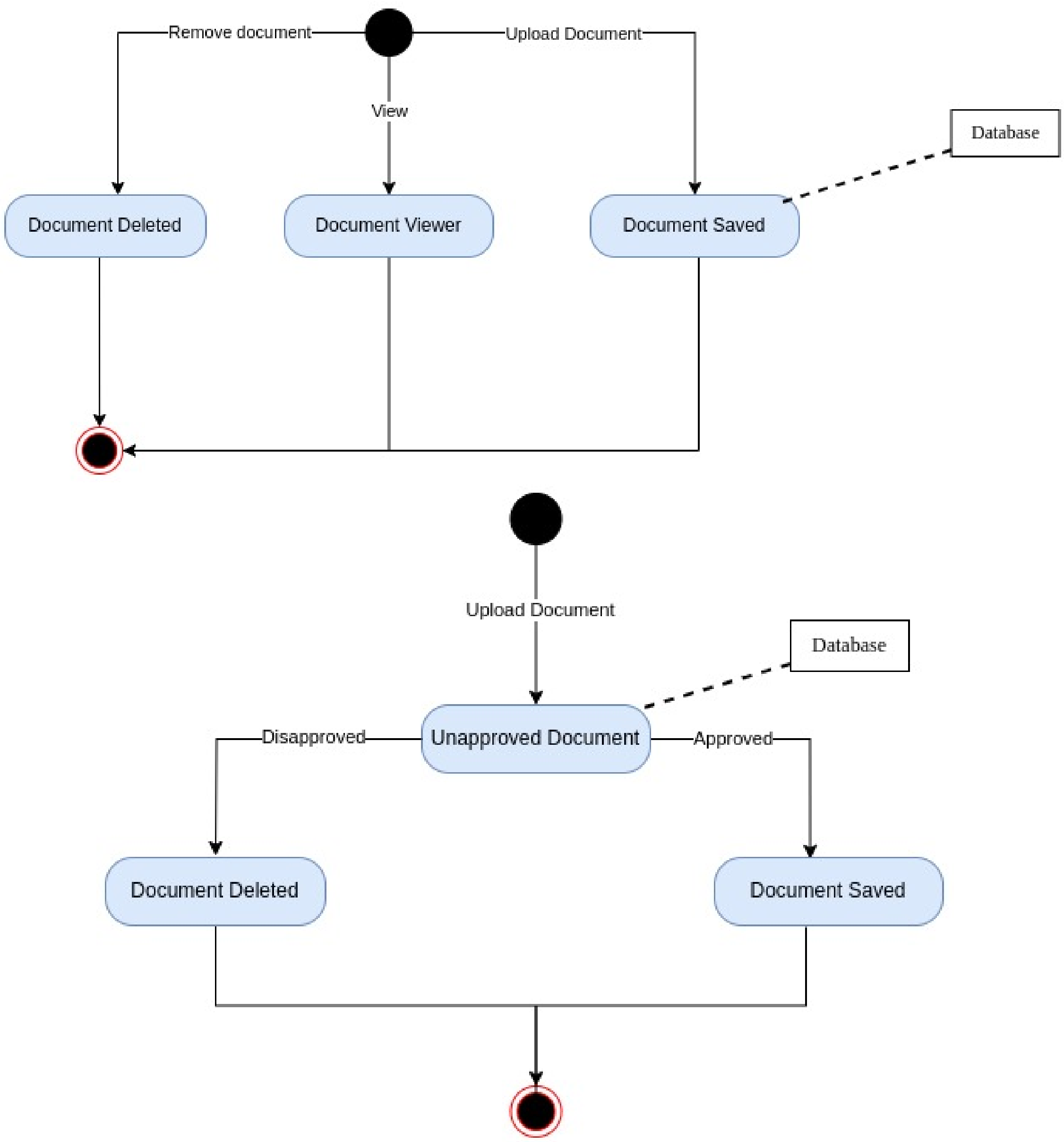

Figure 3.12: State Transition Diagrams: Use-case 1.1 and 1.2

3.3 Data Design

3.3.1 ERD diagram

The following diagram is for MongoDB database which will be used for storing user data, case data, hearing details and keeping track of notifications.

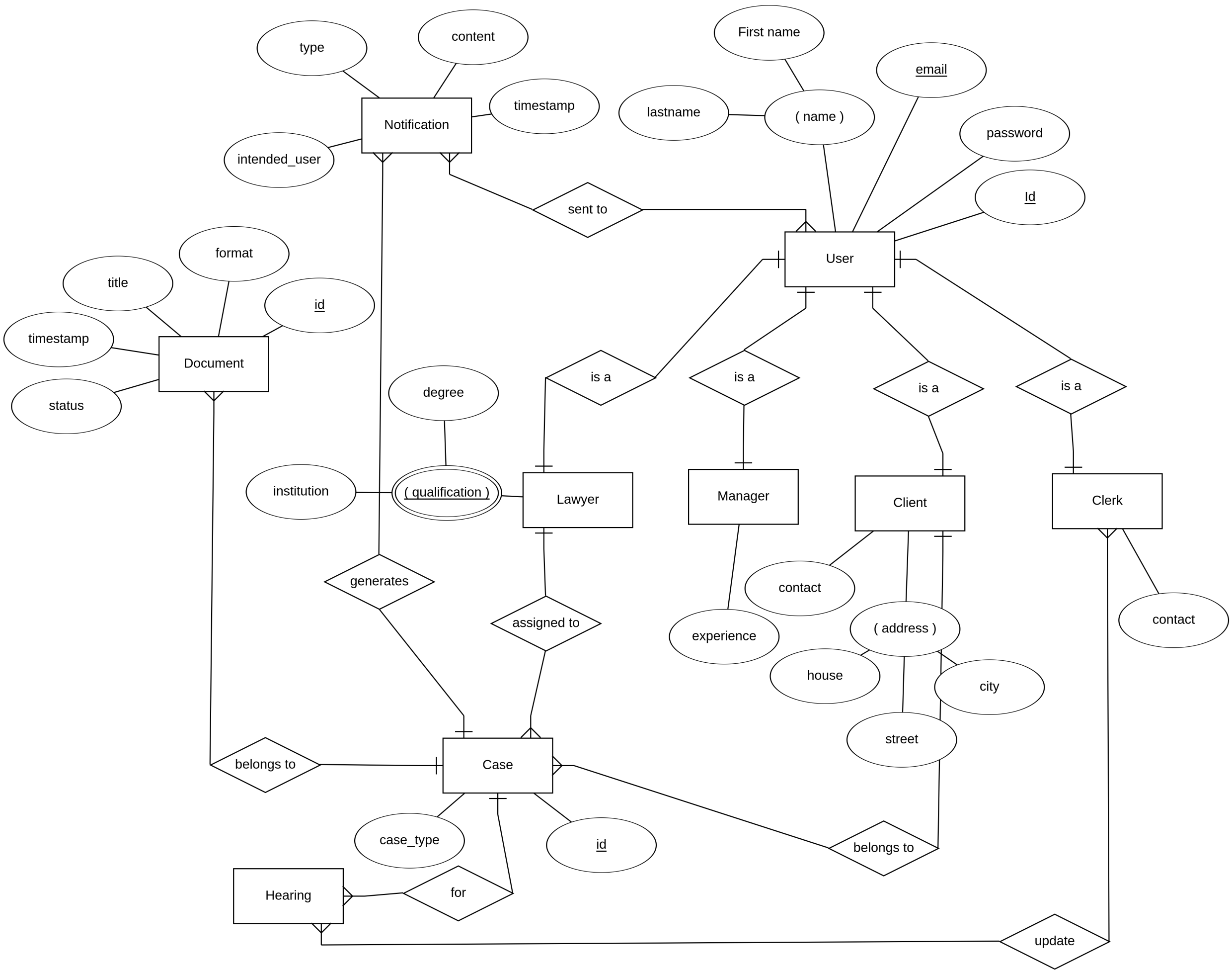

Figure 3.13: CaseLink's Class Diagram

3.3.2 Redis Schema

For the redis database, a detailed key-value pair schema is provided below:

Key: `content:<id>`
Type: Hash
Description: Stores content and its corresponding vector embedding.

Fields

- **id**
  - **Type:** String
  - **Description:** Unique identifier for the content
  - **Required:** Yes
- **content**
  - **Type:** String
  - **Description:** The actual content or document being stored
  - **Required:** Yes
- **embedding**
  - **Type:** Vector
  - **Dimensions:** 128 (adjust based on actual size)
  - **Description:** 128-dimensional vector embedding representing the content
  - **Required:** Yes

Metadata

- **user_id**
  - **Type:** String
  - **Description:** The user who created or uploaded the content
  - **Required:** Yes
- **date**
  - **Type:** Timestamp
  - **Description:** The date when the content was created or added to the database
  - **Default:** Current date

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